Lab Sheet 4 — Linux User Management

Complete questions with answers immediately below each question. Commands are written for common Linux distributions; use sudo only when required.

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distributions; use sudo only when required.

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sudo only when required.

Question 1: Display the current logged-in username, UID, primary GID and supplementary groups.

Answer / Solution:

```
id; id -u; id -g; id -G; groups
```

Question 2: Display the account entry of a test user from /etc/passwd and identify the fields stored in the entry.

Answer / Solution:

```
getent passwd username (fields: name, password placeholder, UID, GID, GECOS, home, shell)
```

Question 3: Display the password/account status of a test user without displaying the actual password.

Answer / Solution:

```
sudo passwd -S username (or sudo chage -l username); do not read /etc/shadow directly for this task.
```

Question 4: Create a new user named labuser1 using useradd and verify that the account has been created.

Answer / Solution:

```
sudo useradd labuser1 && getent passwd labuser1
```

Question 5: Create labuser2 with a home directory and verify the home directory and its ownership.

Answer / Solution:

```
sudo useradd -m labuser2 && ls -ld /home/labuser2
```

Question 6: Create labuser3 with /bin/bash as the login shell and verify the assigned shell.

Answer / Solution:

```
sudo useradd -m -s /bin/bash labuser3 && getent passwd labuser3
```

Question 7: Set a password for labuser1 and verify the password/account status.

Answer / Solution:

```
sudo passwd labuser1; sudo passwd -S labuser1
```

Question 8: Change the password of labuser1 and verify the account status again.

Answer / Solution:

```
sudo passwd labuser1 (enter the new password twice), then sudo passwd -S labuser1
```

Question 9: Use usermod to change labuser2's login shell to /bin/bash and verify the change.

Answer / Solution:

```
sudo usermod -s /bin/bash labuser2 && getent passwd labuser2
```

Question 10: Use usermod to change labuser2's home directory, first without moving the existing contents and then with the option that moves the contents.

Answer / Solution:

```
sudo usermod -d /home/labuser2_new labuser2; then to move contents: sudo usermod -d /home/labuser2_new -m
labuser2 (ensure destination is appropriate first).
```

Question 11: Create a supplementary group named userlabgroup and add labuser1 to it using usermod. Verify the membership.

Answer / Solution:

```
sudo groupadd userlabgroup && sudo usermod -aG userlabgroup labuser1 && id labuser1
```

Question 12: Lock labuser3 and verify that the account is locked. Then unlock it and verify the result.

Answer / Solution:

```
sudo usermod -L labuser3; sudo passwd -S labuser3; sudo usermod -U labuser3; sudo passwd -S labuser3
```

Question 13: Display detailed password-aging information for labuser3 using chage.

Answer / Solution:

```
sudo chage -l labuser3
```

Question 14: Set an account expiration date for labuser3 and verify the expiration information.

Answer / Solution:

```
sudo chage -E 2026-12-31 labuser3 && sudo chage -l labuser3
```

Question 15: Modify labuser1 so that a specified existing group becomes the user's primary group. Verify the result.

Answer / Solution:

```
sudo usermod -g existing_group labuser1 && id labuser1
```

Question 16: Display the users and groups associated with a selected account using id, groups and getent.

Answer / Solution:

```
id username; groups username; getent passwd username; getent group groupname
```

Question 17: Delete labuser1 without removing its home directory. Verify the account entry and home directory separately.

Answer / Solution:

```
sudo userdel labuser1; getent passwd labuser1; ls -ld /home/labuser1 (home should remain unless
separately removed).
```

Question 18: Delete labuser2 and remove its home directory using the appropriate userdel option. Verify the result.

Answer / Solution:

```
sudo userdel -r labuser2; getent passwd labuser2; test ! -e /home/labuser2 && echo "Home removed"
```

Question 19: Create a complete user-management workflow: create a user, set a password, assign a shell, add a supplementary group, lock/unlock the account and verify all settings.

Answer / Solution:

```
sudo useradd -m -s /bin/bash workflowuser && sudo passwd workflowuser && sudo groupadd workflowgroup &&
sudo
usermod -aG workflowgroup workflowuser && sudo usermod -L workflowuser && sudo passwd -S workflowuser &&
sudo
usermod -U workflowuser && id workflowuser
```

Question 20: Clean up all test users and lab-only groups created during this lab. Verify that no unintended system account has been modified.

Answer / Solution:

```
sudo userdel -r labuser1 2>/dev/null; sudo userdel -r labuser2 2>/dev/null; sudo userdel -r labuser3 2>/
dev/null; sudo
groupdel userlabgroup 2>/dev/null; sudo groupdel workflowgroup 2>/dev/null; getent passwd labuser1
labuser2 labuser3
workflowuser; getent group userlabgroup workflowgroup (review each target carefully; do not remove pre-
existing
accounts).
```